

Introduction to Computer Organization and Structure

Subject: Computer Organization and Architecture
Course Code: CS208 Semester: IV Unit: I

Dr. Varun Kumar

varun_kumar@diu.iiitvadodara.ac.in



Indian Institute of Information Technology, Vadodara
International Campus Diu (IIITV-ICD)

Outlines

- 1 Introduction
- 2 Motivation of this Course
- 3 Introduction to ALU
- 4 Instruction representation
- 5 Number of address

Motivation of this course

Key points to remember

- ⇒ Phone price → 300\$ (2022), World fastest computer → 90 Million \$ (1993)
 - ⇒ Innovation in computer design
- ⇒ In late 1970s, the **microprocessor** emerges as a key components for computer design
- ⇒ Two significant changes happens in late 1970s
 - ① Virtual elimination of assembly language programming.
 - ② Vendor independent OS: Ex-UNIX and its clone
- ⇒ In early 1980s → New set of architecture with simple instruction
 - ↓
 - RISC (Reduced instruction set computer)

Key points to remember

- ⇒ RISC had two critical performance techniques
 - ⇒ Instruction level parallelism
 - Initially through pipelining and later through multiple instruction issues
 - ⇒ Use of caches
 - Initially in simple form and later more sophisticated organizations and optimization
- ⇒ The highest-performance microprocessors outperformed the supercomputer of less than 20 year earlier.
- ⇒ Smart cell phone and tablet is preferable over PCs.

Introduction

Computer Architecture

- ⇒ Way of connecting hardware components together to form a computer system.
- ⇒ It is high level abstract for designing a computer.
- ⇒ Help to understand the functionalities of a system.
- ⇒ A programmer can view architecture in terms of
 - ⇒ Instruction formats
 - ⇒ Addressing mode
 - ⇒ Registers
 - ⇒ Data memory

Computer organization

- ⇒ Concerned with the structure and behavior of a computer system as seen by the user.
- ⇒ It deals with the components of a connection in a system.
- ⇒ Organization tells us how exactly all the units in the system are arranged and interconnected.
- ⇒ Organization expresses the realization of architecture.

Key points to remember

- ⇒ Many computer manufacturers offer a family of computer models with same architecture but different organization.
- ⇒ Architecture may span many years but organization changes with the technologies.

Function and structure

Function and structure

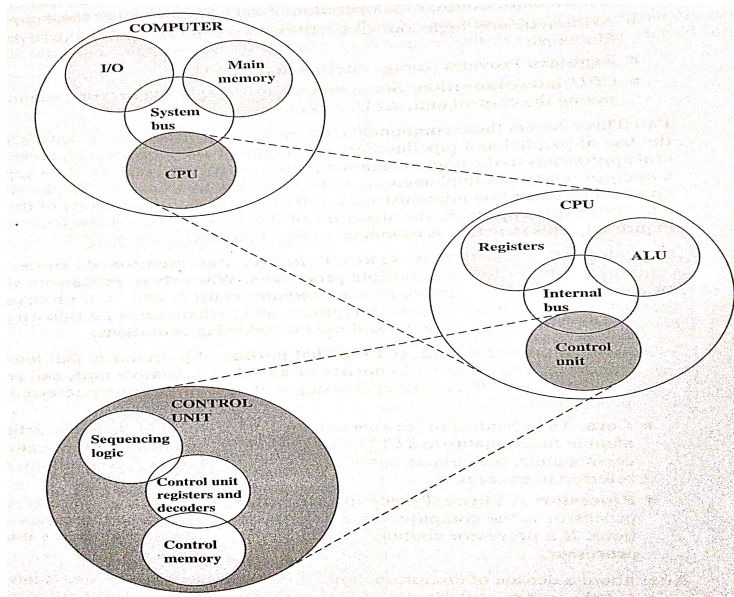
Structure: The way in which the components are interrelated.

- ⇒ Central processing unit → Processor
- ⇒ Main memory → Stores data
- ⇒ I/O → Moves data between computer and outer environment
- ⇒ System interconnection → Mechanism of communication (CPU, Main Memory, I/O)

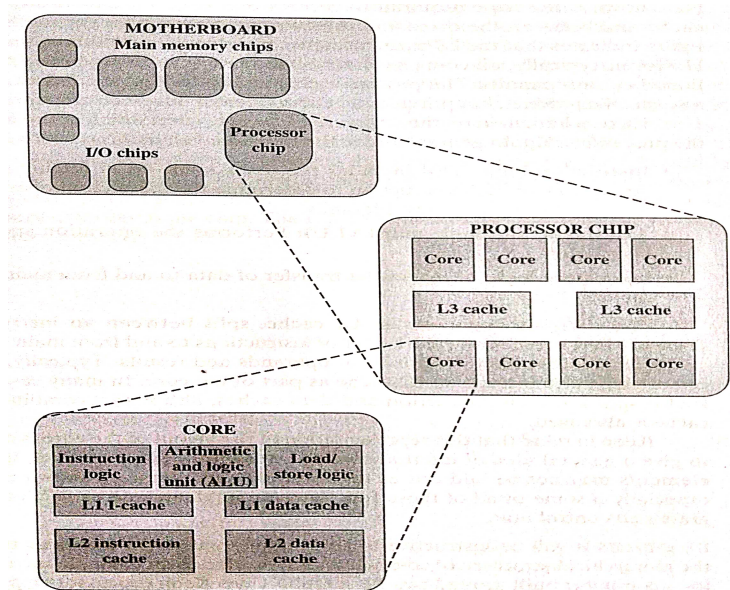
Function: The operation of each component as part of the structure.

- ⇒ Data processing
- ⇒ Data storage
- ⇒ Data movement
- ⇒ Control

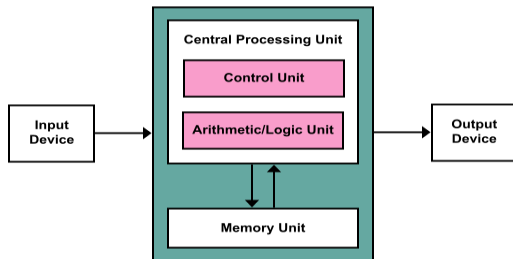
Top Level Structure of Computer



Elements of Multicore Computer



Introduction to ALU



Key points to remember

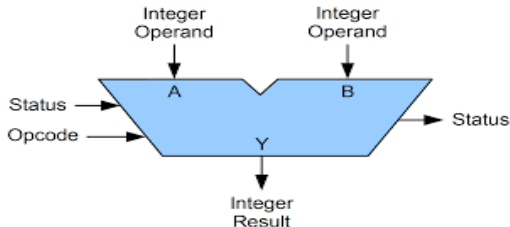
⇒ The ALU unit is capable of performing arithmetic operations on data.

- Addition
- Subtraction
- Multiplication
- Division

ALU ⇒ Logical operation on data

- AND, OR, NOT

Continued–



Key points to remember

- ⇒ The inputs to an ALU are the data to
 - **Operands**
- ⇒ A code indicating the operation to be performed.
 - **Opcode**
 - **ADD, SUB, MUL, DIV** → Arithmetic operation
 - **AND, OR, NOT** → Logical operation

Example:

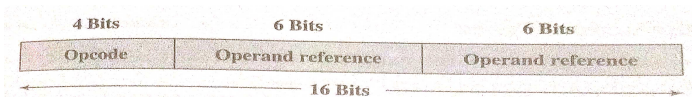
- ⇒ **ADD(A, B)**

Continued—

Key points to remember

- ⇒ **Data:** ALU has three parallel data buses.
 - Two input operands (A and B)
 - Output (Y)
- ⇒ **Opcode:**
 - ⇒ Conveys to the ALU an operation selection code
 - ⇒ The opcode size determines the maximum number of distinct operations the ALU can perform.
 - ⇒ EX- 4-bit opcode → 16 different ALU operations.
- ⇒ **Status:** →
 - ⇒ Convey supplemental information about the result of the current ALU operation.
- ⇒ **Example:**
 - ⇒ Carry-out, Zero, Negative, Overflow, Parity

Instruction representation



Key points to remember

- ⇒ Each instruction is represented by a sequence of bit.
- ⇒ Opcode → *Mnemonics*
 - ADD → Addition
 - SUB → Subtract
 - LOAD → Load data from memory
 - STOR → Store data to memory

Example → ADD R, Y

- ⇒ Add value contained in data location Y to the contents of register R.

Instruction type

- Data processing, Data storage, Data movement, Control

Number of addresses

Program to execute

$$Y = \frac{A - B}{C + (D \times E)}$$

Three-address instruction

Instruction	Comment
SUB Y, A, B	$Y \leftarrow A - B$
MPY T, D, E	$T \leftarrow D \times E$
ADD T, T, C	$T \leftarrow T + C$
DIV Y, Y, T	$Y \leftarrow Y \div T$

Two-address instruction

Instruction	Comment
MOVE Y, A	$Y \leftarrow A$
SUB Y, B	$Y \leftarrow Y - B$
MOVE T, D	$T \leftarrow D$
MPY T, E	$T \leftarrow T \times E$
ADD T, C	$T \leftarrow T + C$
DIV Y, T	$Y \leftarrow Y \div T$

One-address instruction

Instruction	Comment
LOAD D	$AC \leftarrow D$
MPY E	$AC \leftarrow AC \times E$
ADD C	$AC \leftarrow AC + C$
STOR Y	$Y \leftarrow AC$
LOAD A	$AC \leftarrow A$
SUB B	$AC \leftarrow AC - B$
DIV Y	$AC \leftarrow AC \div Y$
STOR Y	$Y \leftarrow AC$

Key points to remember

- ⇒ Three address instruction is not common.
- ⇒ $AC \leftarrow$ Accumulator $\rightarrow$ Processor register

Types of operands

General category of data

① Addresses:

⇒ Calculation performed based on operand address

② Number

⇒ Binary integer or binary fixed point

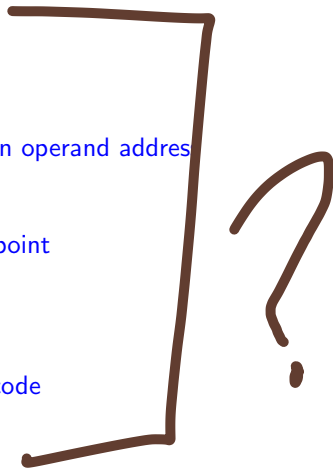
⇒ Binary floating point

⇒ Decimal

③ Character

⇒ Refer → EBCDIC and ASCII code

④ Logical data



Data transfer instruction

Opcode	Operand	Explanation	Example
MOV	Rd, Rs	$Rd = Rs$	MOV A, B
MOV	Rd, M	$Rd = Mc$	MOV A, 2050
MOV	M, Rs	$M = Rs$	MOV 2050, A
MVI	Rd, 8-bit data	$Rd = 8\text{-bit data}$	MVI A, 50
MVI	M, 8-bit data	$M = 8\text{-bit data}$	MVI 2050, 50
LDA	16-bit address	$A = \text{contents at address}$	LDA 2050
STA	16-bit address	$\text{contents at address} = A$	STA 2050
LHLD	16-bit address	directly loads at H & L registers	LHLD 2050
SHLD	16-bit address	directly stores from H & L registers	SHLD 2050
LXI	r.p., 16-bit data	loads the specified register pair with data	LXI H, 3050
LDAX	r.p.	indirectly loads at the accumulator A	LDAX H
STAX	16-bit address	indirectly stores from the accumulator A	STAX 2050